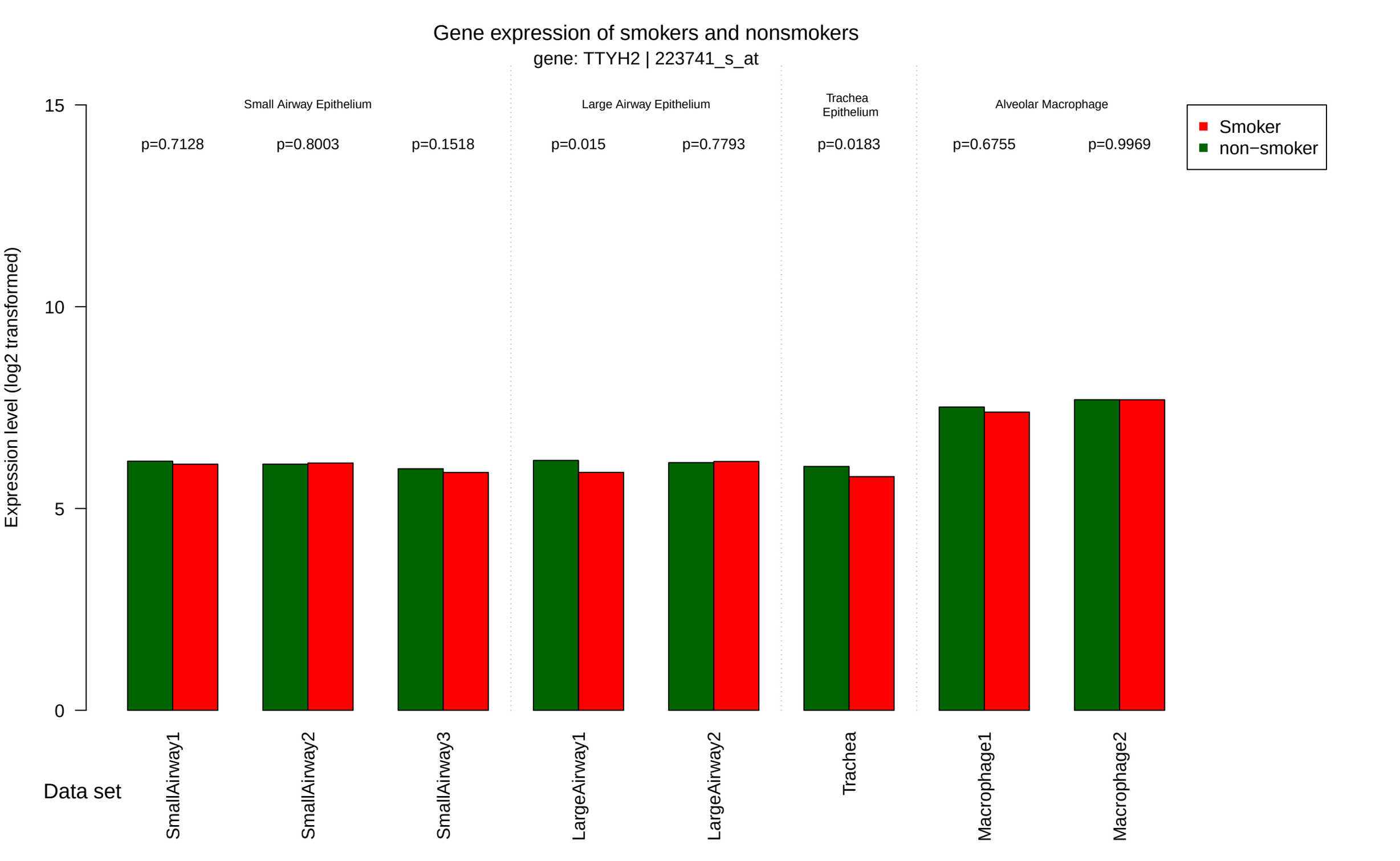
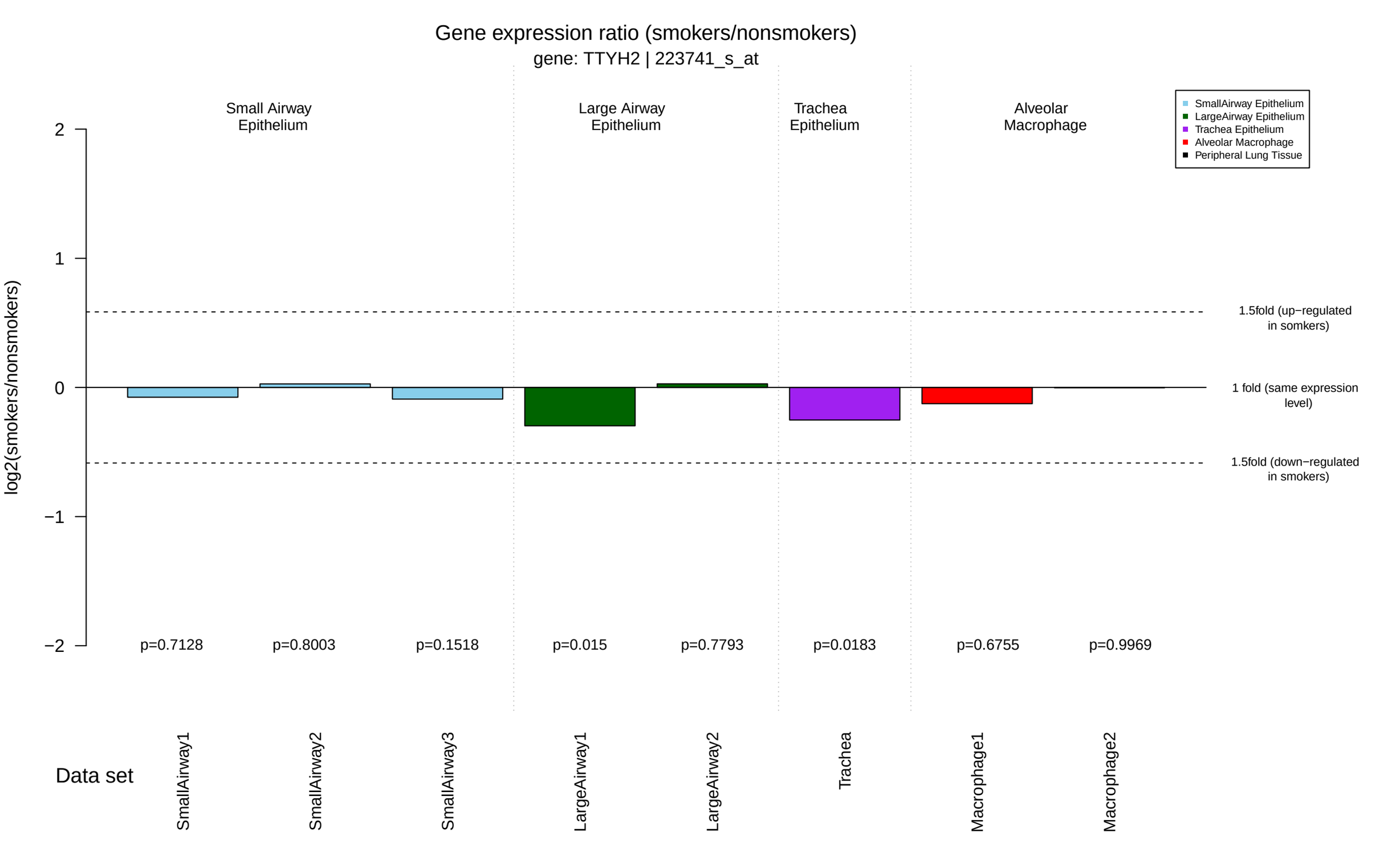
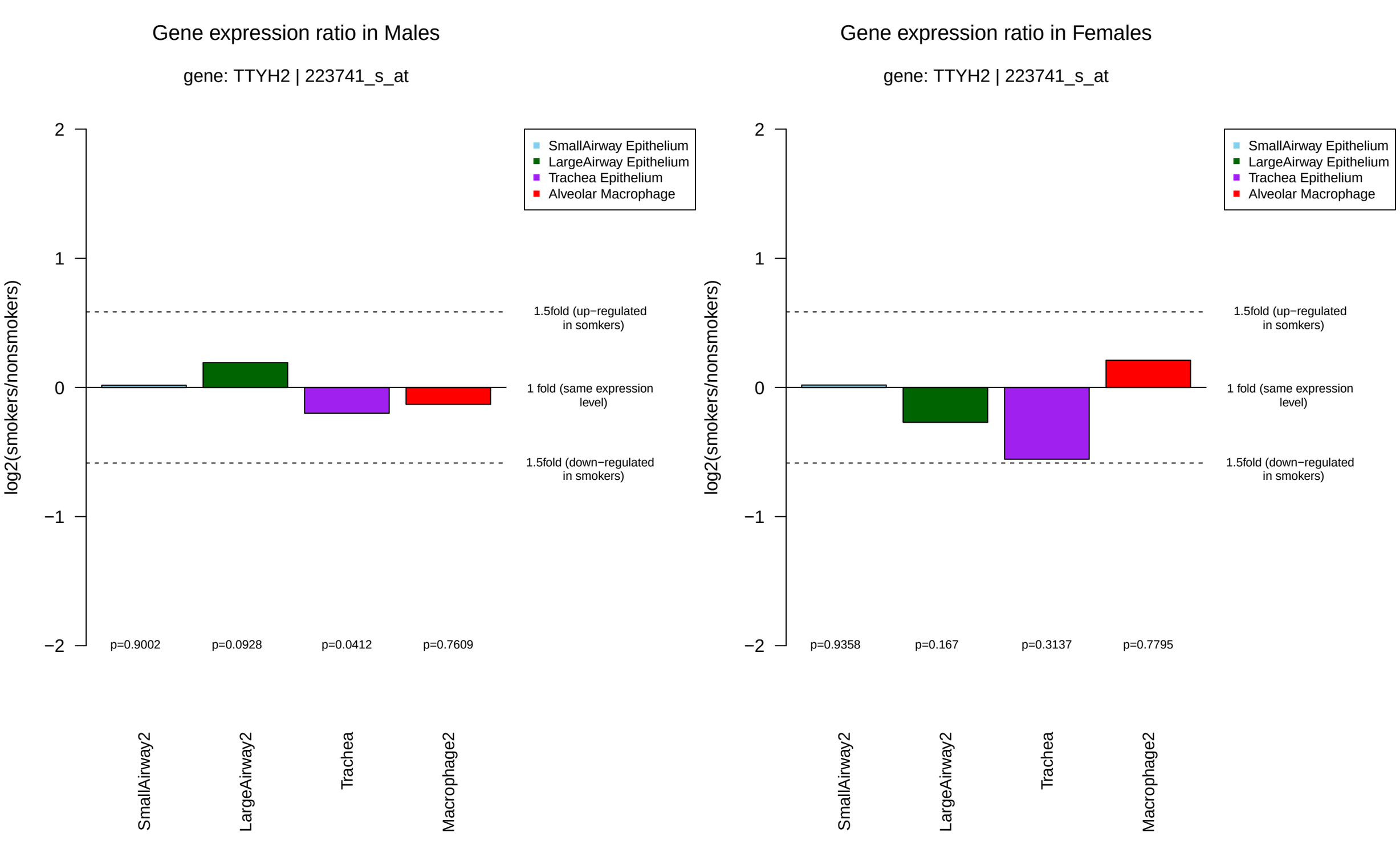
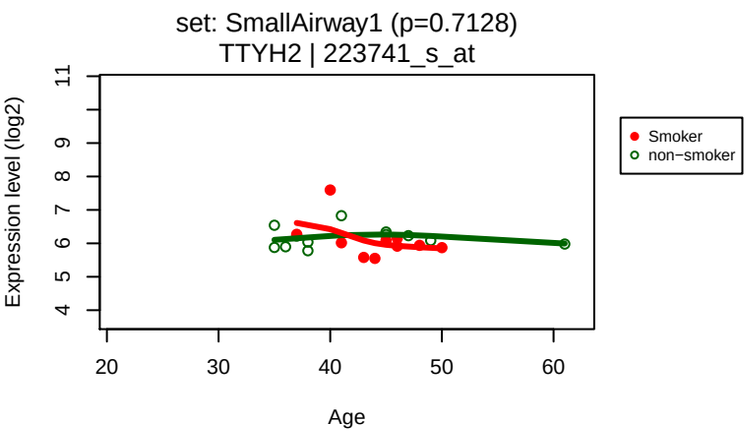
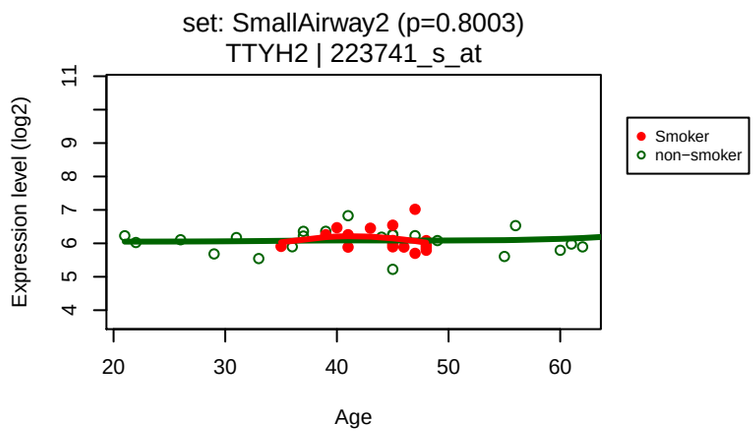
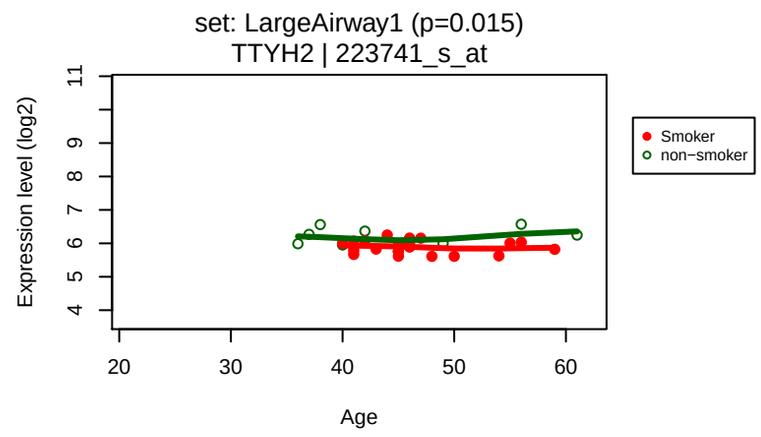

Gene expression of smokers and nonsmokers
gene: TTYH2 | 223741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7128
p=0.8003
p=0.1518
p=0.015
p=0.7793
p=0.0183
p=0.6755
p=0.9969
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

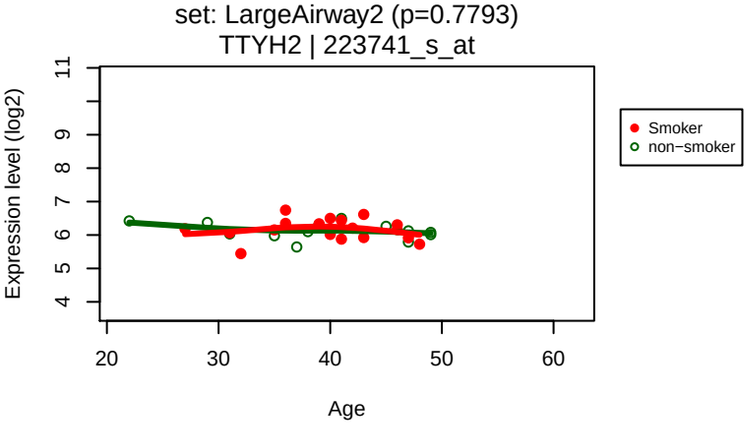

Gene expression ratio (smokers/nonsmokers)
gene: TTYH2 | 223741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7128
p=0.8003
p=0.1518
p=0.015
p=0.7793
p=0.0183
p=0.6755
p=0.9969
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

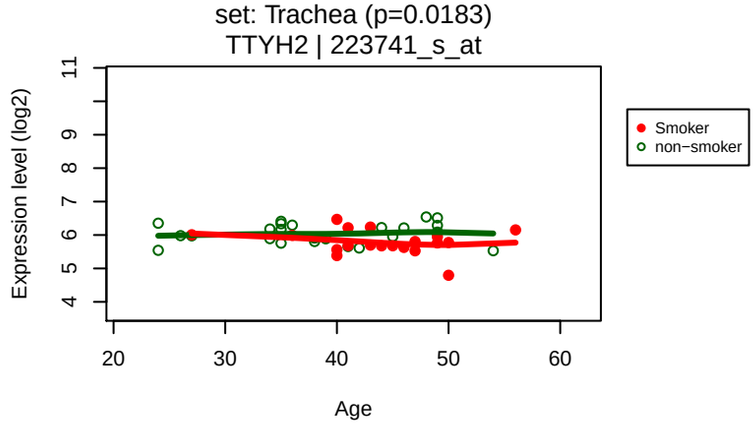

Gene expression ratio in Males
gene: TTYH2 | 223741_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9002
p=0.0928
p=0.0412
p=0.7609
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TTYH2 | 223741_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9358
p=0.167
p=0.3137
p=0.7795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

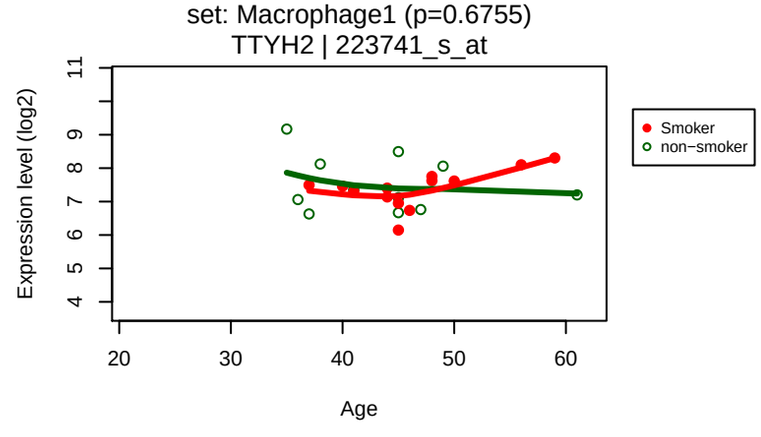

set: SmallAirway1 (p=0.7128)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker

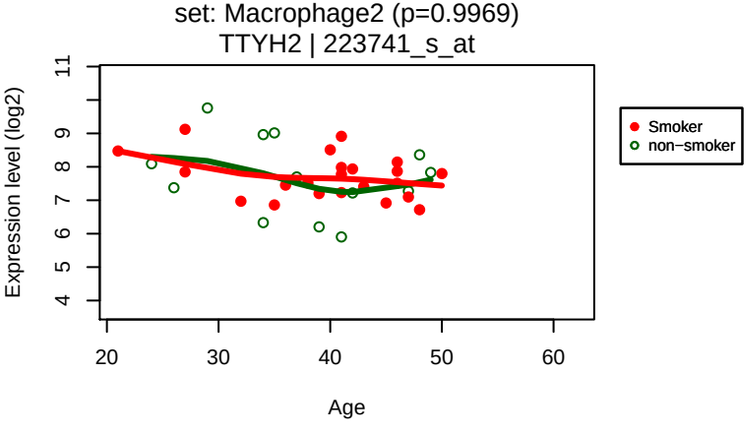

set: SmallAirway2 (p=0.8003)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker

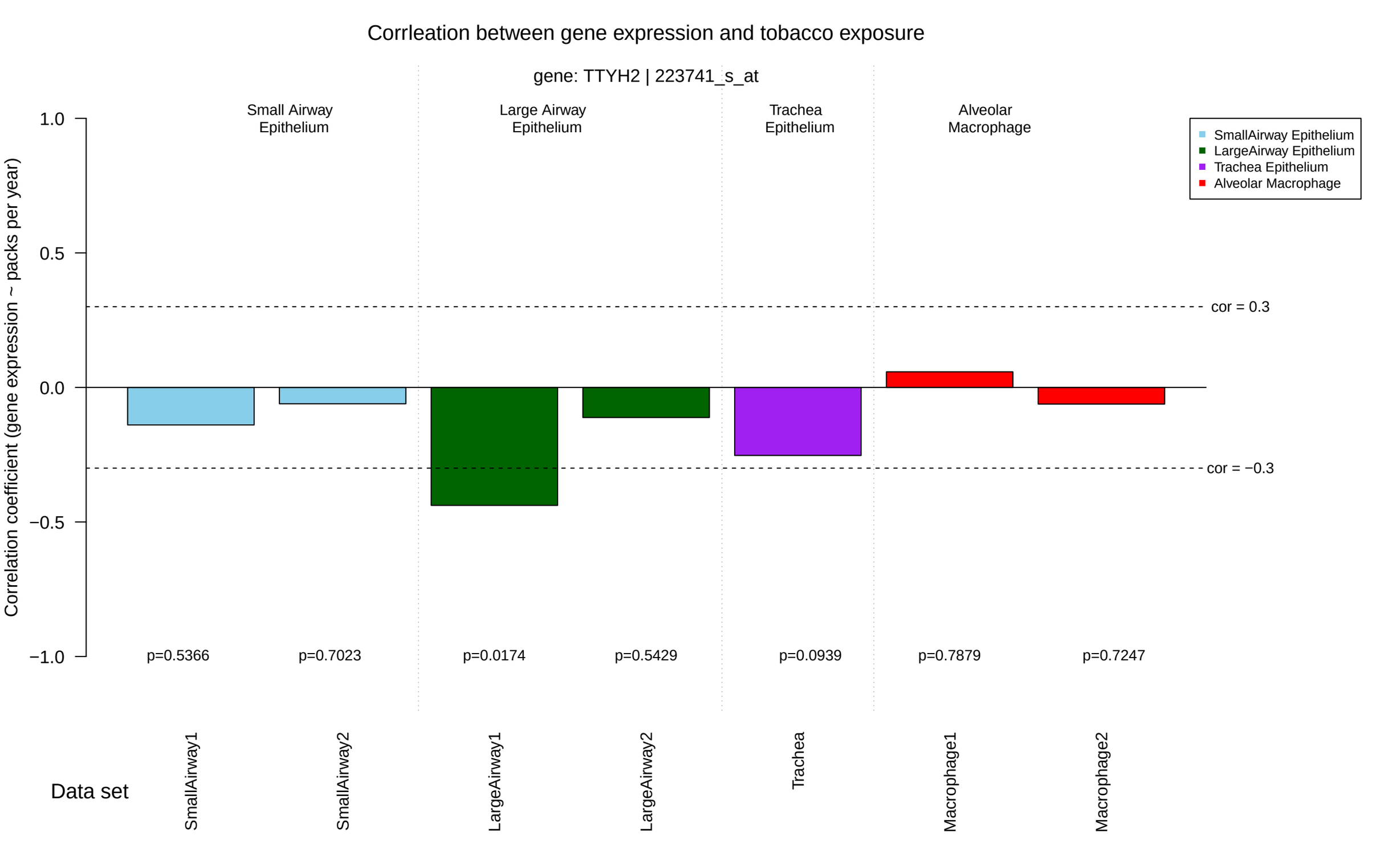

set: LargeAirway1 (p=0.015)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7793)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0183)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6755)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9969)
TTYH2 | 223741_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TTYH2 | 223741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5366
p=0.7023
p=0.0174
p=0.5429
p=0.0939
p=0.7879
p=0.7247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2